Brett Adams

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Education

Honours Computer Science with Software Engineering Specialization Co-op — **Sep 2022 - Present**

University of Windsor — *Windsor, ON*

Skills

Programming Languages	Python | SQL | Java | C++ | C | JavaScript | TypeScript | HTML/CSS | VBA
Data & Cloud	BigQuery | Apache Airflow | Cloud Composer | Azure Databricks | Power BI | MySQL
Technologies	Git | GitLab CI/CD | GitHub Actions | Docker | Node.js | Flask | React | Power Platform

Experience

Data Analyst Intern — **May 2026 - Present**

Geotab — *Oakville, ON*

- Maintain and debug production Apache Airflow DAGs on Cloud Composer that run the SalesOps data pipelines.
- Write and deploy production BigQuery SQL for Revenue Operations tables, including the device status, database dimension and accounts master datasets.
- Ship changes through GitLab CI/CD, from branch development through staging validation to production.
- Carry out live table migrations against fixed infrastructure deadlines, coordinating with dataset owners.

Data Management Associate, Co-op — **Jan 2025 - Aug 2025**

Ontario Ministry of Transportation — *St. Catharines, ON*

- Built Power BI dashboards on Azure Databricks and SQL that replaced manual reporting across business units.
- Automated legacy workflows with SQL queries and VBA scripts, cutting processing time from hours to minutes.
- Led the rollout of the internal Data Broker Tool, onboarding teams and standardizing the data vocabulary.

Digital Solutions, Co-op — **Apr 2024 - Sep 2024**

CIMCO Refrigeration — *Burlington, ON*

- Built Flask web applications that deliver real-time data and alarm information to NHL arena operations teams, including a thermal imagery viewer used by arena technicians.
- Wrote a Python pipeline that aggregates data from two APIs into structured JSON for the arena interface.
- Automated controller data collection for CAG engineers with a Python tool, replacing manual data entry.
- Designed and ran hands-on Python training sessions for over 20 employees.

Digital Solutions — **Apr 2023 - Sep 2023**

CIMCO Refrigeration — *Burlington, ON*

- Built Power Platform tools adopted by several departments to automate repetitive manual workflows.
- Automated SharePoint document routing and approvals, cutting turnaround time between departments.

Notable Projects

mcp-surface — **Jul 2026 - Present**

TypeScript CLI and GitHub Action for testing Model Context Protocol servers — *Independent*

- Published on npm. Connects to an MCP server over stdio, Streamable HTTP or SSE the way a real client does, and reports the tools, resources and prompts it actually exposes.
- Runs 19 checks against what the server returns, such as invalid JSON Schemas, unsatisfiable required fields and duplicate tool names.
- Snapshots the tool surface and fails CI when it changes unexpectedly; also ships as a GitHub Action.

MCP Servers for Claude Code — **Jul 2026 - Present**

Five Node.js servers exposing 69 tools to AI assistants — *Independent*

- Covers Google Workspace, Reddit, Yahoo Fantasy, Chess.com and three Supercell game APIs, handling OAuth 2.0 and API-key auth.
- The Google Workspace server (23 tools) sends email and manages Calendar, Drive, Sheets and YouTube through a self-registered OAuth client that requests only non-restricted scopes.
- Each server has a test suite that runs with no network access or credentials, and CI on Node 20 and 22.

MazeFinder — **Mar 2024 - Sep 2024**

C++ application for solving 20x20 mazes using genetic algorithms — *Independent*

- Evolves paths through 20x20 mazes with a genetic algorithm using dynamic mutation and crossover strategies.
- Parallelized genome fitness evaluation with a custom thread pool.
- Stores mazes as 2D vectors so new maze configurations load without code changes.